

Recompute, Don't Trust: A Sealed Record-Replay Benchmark of Which Verifier Catches a False Numeric Self-Report

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Abstract

LLM agents report their own numeric results, and autonomous pipelines increasingly consume those self-reports unchecked. On a mechanically checkable task — a model reads a caller-owned dataset and states how many records it contains — we record real transcripts once and replay them deterministically, so the verifier is the only free variable and ground truth is sealed by cryptographic commitment before recording. On a 16-error judged corpus the verifiers separate cleanly: a schema guard catches 0/16; a cross-judge consistency vote catches 13/16 but abstains on exactly the near-misses; tool-less LLM judges catch 47/48, their sole failure a judge accepting an error identical to its own miscount; per-step recomputation catches 16/16 — and 160/160 across the full corpus — with zero false rejections on more than one hundred correct claims [E-EXPT-010, E-EXPT-009, E-EXPT-011, E-EXPT-018]. In six recorded three-step chains an early miscount propagated arithmetically-coherently into the model's own derived claim, invisible to self-consistency, while per-step recomputation halted every chain at the first wrong step [E-EXPT-003, E-EXPT-004]. The capability pattern replicates across a second vendor family, and neither rate saturation below the frontier nor an 8K-token reasoning budget changes the verdict ordering [E-EXPT-018, E-EXPT-012]. The contributions are the sealed record-replay instrument, its honesty machinery, and these measured differentials on identical recorded claims.

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1. Introduction

An agent finishes a step and states a result. In an autonomous pipeline the number is consumed directly — the next stage proceeds, a report ships — because rechecking every claim is expensive and the claim looks right. The stakes are measured elsewhere: research agents fabricate or invalidate experimental results in a large majority of cases (Chen et al., 2025), fabricated papers pass automated LLM review at rates up to 82% (Jiang et al., 2025), and a majority of manuscripts from an autonomous scientist contained hallucinated numbers (Beel et al., 2025). These are failures of the claim channel: what an agent reports about its own work is wrong, and nothing downstream checks it.

We study a deliberately simple instance — a model reads a caller-owned dataset and claims how many records it contains — so the truth is recomputable and an error is unambiguous. On this channel we measure which verifier catches a false self-report, and what happens when a false report is chained. Our claims (C1–C8, mapped to evidence in Appendix B): a measured verifier hierarchy on identical recorded claims (C3); blast-radius containment of coherently-propagating chain errors (C8); genuine spontaneous errors that are plausible near-misses invisible to format checks (C1, C2); corroborating instructed probes (C5); an envelope-conformance failure axis distinct from value error (C7); a capability observation spanning three vendor families (C6); and the harness itself with honesty enforced structurally (C4).

Not new, and not claimed: that a schema guard cannot inspect a value; that recomputation catches recomputable errors; that step-level verification beats outcome-level checking, which is the process-verifier result (Lightman et al., 2023) (Cobbe et al., 2021). New here: the measured differential among a schema guard, transplanted consistency verifiers, tool-less judges, and recomputation on the same recorded claims; the recorded mechanism showing why agreement-based checks fail on chains; and a sealed record-replay protocol whose honesty machinery is validated by self-caught confounds.

2. Related Work

Agent benchmarks score task success, not claim fidelity (Mialon et al., 2023) (Yao et al., 2024) (Backlund & Petersson, 2025); autonomy evaluations document overclaiming (METR, 2026); the fabrication line quantifies how often false numbers ship (Chen et al., 2025) (Beel et al., 2025). We isolate the claim channel and attach the verifier comparison they lack. Feedback-loop methods — self-refinement (Madaan et al., 2023), tool-interactive critique (Gou et al., 2023), sampling-agreement detection (Manakul et al., 2023), calibration (Kadavath et al., 2022), answer self-consistency (Wang et al., 2022) — feed the acceptance decision we measure; our chain result shows why agreement over a chain’s own numbers cannot audit it. Process-level verification is established (Lightman et al., 2023) (Cobbe et al., 2021); our per-step recomputation is its deterministic, training-free analogue for mechanically checkable claims, and our delta is the measured differential plus the instrument, not the principle. Guardrails frameworks validate typed output (Guardrails AI, 2023); LLM-judge failure modes are documented (Zheng et al., 2023); we give both a concrete mechanistic instance on the same corpus. Design-by-contract (Meyer, 1992) and policy-as-code (Open Policy Agent contributors, 2016) supply the obligation vocabulary; neither measures which runtime check catches a false claim.

3. Design

The harness records a real model transcript once — the model reads an inlined dataset tool-lessly and states a claim — and replays it deterministically offline through a governed path, so the recorded output is held fixed while the verifier varies (Figures 1 and 2). A claim never advances itself: only an accepted verdict does, and for chains the same gate runs between steps. Honesty is structural (C4): unrecorded placeholders refuse replay; synthetic fixtures are non-citable; parse-failures are recorded data, never retried; ground-truth counts for every new arm were sealed by hash commitment before any transcript was recorded (a sha256 commitment revealed only after all recording completed, making blinding third-party auditable), pinned in tests [E-EXPT-001, E-XLOG-001]. The protocol has twice caught its own confounds — a shortcut-compressible dataset and a non-replicating scale effect — because every prompt is preserved (Appendix A). The reference implementation passes 225 tests [E-XLOG-001], and produces identical verdicts across an in-process substrate and a real Agent Client Protocol session on all five conditions [E-EXPT-002] (Zed Indus-

tries, 2025).

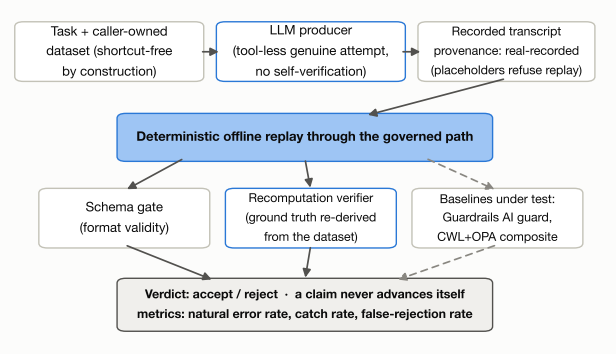


Figure 1. F1: Benchmark harness — record once, replay deterministically through layered gates and the baselines under test.

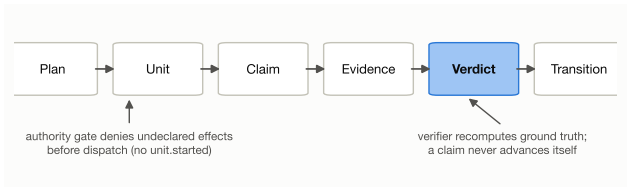


Figure 2. F2: Governed run pipeline — a claim advances only as Plan, Unit, Claim, Evidence, Verdict, Transition; the verdict gate sits between every step.

4. Evaluation

A measured detection hierarchy (C3). We score four verifier tiers on the same judged corpus: the 4 scale2 near-miss errors plus the 12 smallest-|error| capability-ladder claims, 16 natural errors in all, each with 3 recorded judge verdicts [E-EXPT-010]. A structural guard — a real Guardrails AI validator mirroring the evidence contract — accepts every schema-valid claim: 0/16 caught, because every recorded error is a well-formed integer; the same guard does catch the three pure format violations (C2, C7) [E-EXPT-008]. The cross-judge majority-recount vote catches 13/16 with 3 abstentions — and the abstentions fall exactly on the scale2 near-misses, where the judges’ own recounts disagree: agreement-based verification works on gross errors and collapses precisely where the task is hard; the ladder 12/12 is moreover decided by shared within-dataset recounts (Appendix C), not independent per-claim evidence, so it reads alongside the scale2 1/4 [E-EXPT-009, E-EXPT-006]. Three tool-less LLM judges catch 47/48 combined (15/16, 16/16, 16/16), including every off-by-one in the ladder set; the tier’s sole failure across all 48 judgments is recount-coincidence — a judge accepting 1552 against a true 1551 because its own recount was identically wrong — and judges

reach correct verdicts despite inaccurate recounts in 5 of 27 scale2 judgments, so a right verdict is not evidence of a right count — and conversely, the smallest judge tier’s own recount of the large dataset was wrong (768 against a true 754) yet it still rejected the claimed 767, because 767 differed from its own count [E-EXPT-010, E-EXPT-011]. Ground-truth recomputation catches 16/16 judged and 160/160 across the combined corpus, with zero false rejections [E-EXPT-018]. Its completeness on a recomputable task is a structural guarantee, conceded; the empirical content is the differential above it.

Table note. N is the base of each caught fraction: guard, vote, and recomputation score once per error (16); the judges pool three tiers (48 judgments). False-rejection denominators are the correct claims each method judged: 5 scale2 claims per method, 15 across the judge tiers, and for recomputation the 101 correct cross-vendor claims.

Chains: coherent propagation, first-step containment (C8). Six recorded three-step chains (count $\rightarrow$ sum $\rightarrow$ derive, later steps consuming the model’s own earlier claims) carried an erroneous first step — one frontier chain and five scripted local chains across three model sizes. In all six the early miscount propagated arithmetically-coherently into the model’s own derived final claim, so a self-consistency check over the chain’s own numbers would ratify it; per-step recomputation rejected the count and halted every chain at the first wrong step, and the two clean frontier chains were accepted end-to-end [E-EXPT-003, E-EXPT-004]. One recorded instance makes the mechanism concrete: a mid-size local model claimed 453 records against a true 273, summed its second dataset to 9,848, and derived 10,301 — exactly $453 + 9,848$ — so only a verifier holding ground truth external to the chain can break the agreement [E-EXPT-003]. Two local chains additionally mis-added their own numbers — a further inconsistency axis (Appendix C).

Errors are near-misses; the pattern replicates across vendors (C1, C2, C6). Among three frontier models of one family the recorded natural error rate orders by capability — 0.15, 0.31, 0.54 (2/13, 4/13, 7/13) — but no pairwise difference is significant at $N=13$ (Fisher $p = 0.65/0.43/0.097$) and the apparent scale effect did not survive de-confounding: two original large datasets showed elevated error (6/12, 7/12) yet four structurally comparable independent datasets at the same scales produced 0/12, leaving dataset identity and instruction wording inseparable; we report the non-replication as the instrument working [E-EXPT-014, E-EXPT-015]. In a second vendor family under a drift-

guarded identical task (reasoning=high), two capable tiers err never (0/40, 0/40) and the weaker tier errs on 19/40 — with error magnitudes that are near-misses (mean abs err 1.26, median 1.0, max 3 — never gross) [E-EXPT-018]. Below the frontier, rate saturates: three local qwen3 models err on every episode (40/40 each) while magnitude still shrinks with scale (mean $297 \rightarrow 115 \rightarrow 102$) and off-by-one errors appear only at the larger sizes; an 8K-token reasoning budget does not rescue the smallest model (32/40 no answer; the 8 answers all wrong, mean $|\text{err}| = 531$) [E-EXPT-012, E-EXPT-013]. Figure 3 shows the local panels; protocols across arms are disclosed and non-commensurable, so we compare by raw rate only (Appendix A).

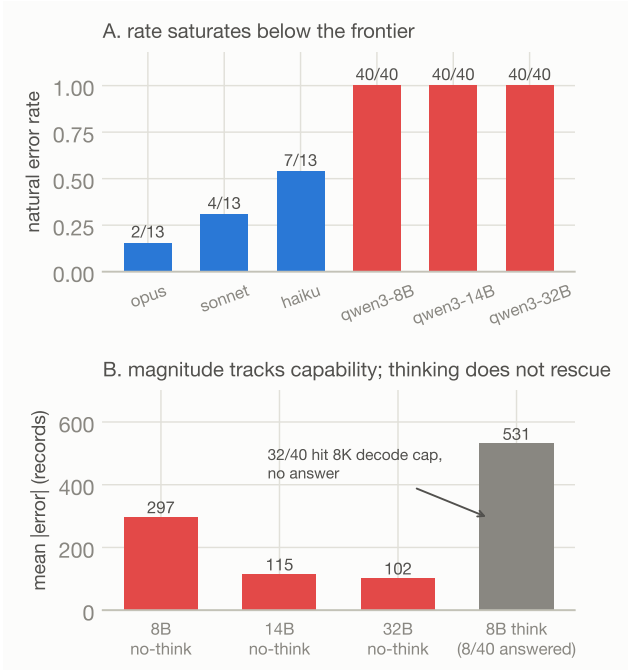


Figure 3. F3: Local family capability panels: error rate saturates at 1.00 across 8B/14B/32B while mean absolute error shrinks with scale (297/115/102), and the 8B think arm (8K-token budget) fails to terminate or answers with larger error.

Instructed probes corroborate (C5). Replaying 24 recorded instructed episodes across three task families: every true claim accepted (9/9), every injected false claim rejected (15/15), including borderline off-by-ones caught only by recomputation (Figure 4) [E-EXPT-007]. Small models additionally fail claim-envelope conformance — a failure axis distinct from value error (C7) [E-EXPT-001].

Verifier	Caught	N (errors)	False rejections
Schema guard (Guardrails AI)	0/16	16	0/5
Cross-judge consistency	13/16 (ladder 12/12, near-miss 1/4)	16	0/5
Single tool-less LLM judges	47/48	48 judgments	0/15
Ground-truth recomputation	16/16 (160/160 corpus)	16	0 (incl. 0/101)

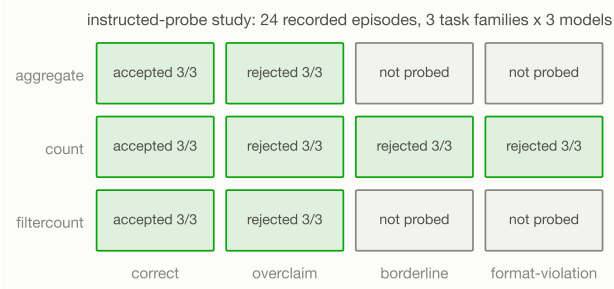


Figure 4. F5: Instructed-probe outcome matrix (24 episodes, three task families): every true claim accepted, every injected false claim rejected.

5. Discussion and Limitations

Black-box ceiling, white-box next tier. Every tier we measure is black-box: schema, agreement, judges, and recomputation read only outputs. The recount-coincidence miss and the self-ratifying chains are behavioral evidence that output-side reading cannot audit a claim; auditing requires ground-truth recomputation or inspection of internal states — the direction instantiated by global-workspace interpretability (Team, 2026). Our corpus is a natural testbed: every claim is a recorded open-weight transcript with sealed ground truth, so a white-box claim-verifier can be scored against identical claims and slotted into this hierarchy. Whether internal signals discriminate is the open question the corpus enables.

Practical reading. For a deployer the differential translates directly: validate format with a guard (necessary but insufficient); do not substitute agreement for truth — votes abstain or false-flag exactly where verification matters most; treat a judge as a strong screen whose verdicts cannot be audited from its own recounts; and wherever a claim is mechanically checkable, recompute it per step — the only tier here that is simultaneously complete, auditable, and chain-containing.

Bounded scope. Every result is a statement about this corpus: 16 judged errors, 6 early-error chains, one recomputable task family — modest but replayable. Two tiers are analytic endpoints, conceded. The consistency verifiers are derived from recorded data, with

within-dataset recount sharing disclosed (Appendix C). The frontier ordering is suggestive, not significant; the cross-vendor arm uses a different generation backend with drift-guarded task identity (Appendix A). No general capability law is claimed from three families on one task.

6. Conclusion

Agents report their own results and pipelines increasingly trust the report. On identical recorded numeric self-reports, verification methods separate into a measured hierarchy: format guards catch none of the value errors; agreement-based checks work on gross errors and abstain exactly on near-misses; tool-less judges catch almost everything but are unauditable in principle — a correct verdict is not evidence of a correct count; and only per-step recomputation is complete and auditable, containing chain errors before they compound in all six recorded early-error chains. For any system consuming an agent’s checkable numeric self-report, the practical rule is the title’s: recompute, don’t trust.

7. Reproducibility Statement

Every quantitative statement carries an [E...] identifier resolving into the released research bundle, and every result is reproduced by replaying the recorded episodes through the released harness; ladder and cross-vendor ground truths were hash-committed before recording. Appendix A gives the reproduction map and runner invocations.

8. Impact Statement

This work is defensive: it measures how often LLM agents misreport their own numeric results and shows that only ground-truth recomputation reliably catches such errors and contains their propagation through multi-step pipelines. The benchmark uses synthetic counting datasets and recorded transcripts, with no personal or sensitive data. A residual risk is over-generalization: results are scoped to one recomputable task family and a fixed model set.

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A. Protocols, Confounds, and Reproduction Map

Datasets are shortcut-free by construction (random non-sequential ids, irregular group sizes, generator assertions). Two self-caught confounds: an early dataset family admitted a count-revealing shortcut (regenerated; the suspicious 9/9 documented), and an apparent frontier scale effect did not replicate on four structurally comparable independent datasets (0/12 vs 6/12, 7/12), leaving dataset identity and instruction wording inseparable [E-EXPT-014, E-EXPT-015]. Arms and protocols: frontier grids are single recorded transcripts per cell; the local arms are temperature-varied harvester generations (seed and temperature

per episode, num_ctx asserted untruncated) [E-EXPT-012, E-EXPT-013]; the cross-vendor arm swaps only the generation backend, with the task text byte-identical under a drift-guard test [E-EXPT-018]. Ground truths for the ladder and cross-vendor arms were sealed by sha256 commitment over salted canonical JSON before any recording; the preimage is published at reveal and the commitment is pinned in tests [E-EXPT-001, E-XLOG-001]. Reproduction: each artifact in the bundle regenerates from its runner over the committed fixtures — capability [E-EXPT-012, E-EXPT-013, E-EXPT-018, E-EXPT-014], judges [E-EXPT-010, E-EXPT-011], consistency [E-EXPT-009, E-EXPT-006], baselines [E-EXPT-008, E-EXPT-005], chains [E-EXPT-003, E-EXPT-004], probes [E-EXPT-007], substrate portability [E-EXPT-002]; source and suite at [E-SRC-001, E-XLOG-001].

Runner invocations (repository root; committed fixtures only): the full suite via pytest (225 tests); natural/power study, judge study, ladder judge study, ladder consistency study, and consistency study as python modules under contractplane.experimental; chain scoring via examples/governed-run/chain_study.py (frontier dir and -dir episodes/study-chain-gpu); the 8B baseline via local_study_runner.py; the ladder and cross-vendor arms via family_study_runner.py with -family local-gpu and -family openai; substrate agreement via acp_substrate_study.py; seal verification via the sealing module given the committed seal and revealed preimage.

The governed path additionally reproduces its verdicts across an in-process substrate and a real Agent Client Protocol session (five of five conditions agree) [E-EXPT-002], and a composite CWL+OPA baseline comparison locates the enforcement-representability delta [E-EXPT-005].

B. Claim Codebook

C. Judge and Chain Corpus Construction

The judged corpus joins the 4 scale2 near-miss errors (three judge tiers, 27 judgments, recount accuracy 21/27) [E-EXPT-011] with the 12 smallest-|error| ladder claims (36 fresh judge verdicts; per-tier recount accuracy 12/12, 12/12, 11/12 — one judge’s own recount of the large dataset was 768 against a true 754 yet its verdicts stayed correct) [E-EXPT-010]. Ladder judges made one genuine tool-less pass per dataset, so within-dataset votes share each judge’s recount; the derivation exposes this structurally [E-EXPT-009]. One judge tier disclosed a line-count (wc -l) protocol deviation, preserved verbatim in the study notes; line counts

carry no record-composition information. Chains: the frontier chain and five scripted local chains follow the identical three-step protocol, later prompts embedding the model’s own earlier claims verbatim; one local chain emitted an unevaluated expression at the derive step and was recorded as an honest abort, excluded from scoring; two local chains mis-added their own two numbers, a further self-inconsistency axis [E-EXPT-003].

D. Full Result Tables

Natural-error arms, replayed through the identical governed path. Frontier producers are named by relative tier for review anonymity. Recomputation caught every error in every arm with zero false rejections (160/160 combined) [E-EXPT-014, E-EXPT-018, E-EXPT-013, E-EXPT-012].

Recomputation’s 160 combined catches decompose as 88 from the GPU ladder, 40 from the separate 8B run, 13 from the frontier power study, and 19 from the cross-vendor family; the zero-false-rejection evidence with genuine correct claims is 0/15 judged-correct, 0/26 correct frontier episodes, and 0/101 correct cross-vendor claims. Conditions: frontier = one recorded transcript per grid cell; cross-vendor = reasoning high, N=20 each on datasets f and g, zero parse failures; local = temperature-varied harvester, no-think; think arm additionally hit the decode cap with no answer on 32 of 40 attempts.

Judge tiers on the judged corpus (judge models by relative tier) and the consistency vote [E-EXPT-010, E-EXPT-011, E-EXPT-009, E-EXPT-006]:

Every judge caught all three ladder off-by-one claims; false rejections 0/5 per judge on the scale2 correct set. Judge L’s sole miss is the recount-coincidence case (accepting 1552 against a true 1551). Ladder recount accuracy: 12/12, 12/12, 11/12 — Judge S recounted the large dataset as 768 against a true 754 yet its verdicts stayed correct. Scale2 recount accuracy over 27 judgments: 21/27, with 5 of 27 correct verdicts carried by inaccurate recounts; on the ladder set the tiers recounted accurately in 35/36 judgments and reached correct verdicts in 36/36. All three vote abstentions fall on scale2 near-misses where the judges’ own recounts disagree.

All eight recorded chains (truths: count 273, sum 38808, derived 39081) [E-EXPT-003, E-EXPT-004]:

Every halted chain’s derived claim consumed its own wrong count; five of six error chains are internally consistent with their own arithmetic (the 32B(2) and 8B(2) chains additionally mis-add their own numbers:

Claim	Statement (abridged)	Evidence
C1	Genuine spontaneous errors; the apparent frontier scale effect did not replicate	[E-EXPT-014, E-EXPT-015]
C2	Errors are plausible near-misses invisible to format checks	[E-EXPT-008, E-EXPT-015, E-EXPT-018]
C3	Measured verifier hierarchy on the 16-error judged corpus	[E-EXPT-010, E-EXPT-009, E-EXPT-011, E-EXPT-006]
C4	Honesty machinery: placeholder refusal, sealed truths, parse-failure-as-data	[E-SRC-001, E-XLOG-001, E-EXPT-001]
C5	Instructed probes: 9/9 true accepted, 15/15 false rejected	[E-EXPT-007]
C6	Capability observation across three vendor families	[E-EXPT-018, E-EXPT-012, E-EXPT-013, E-EXPT-014]
C7	Claim-envelope conformance failures distinct from value error	[E-EXPT-001]
C8	Chain propagation and first-step containment (6/6 early-error chains)	[E-EXPT-003, E-EXPT-004]

Arm / model	N	Errors	Rate	Abs err
Frontier large	13	2	0.154	near-miss
Frontier mid	13	4	0.308	near-miss
Frontier small	13	7	0.538	near-miss
gpt-5.6-sol	40	0	0.000	—
gpt-5.6-terra	40	0	0.000	—
gpt-5.6-luna	40	19	0.475	mean 1.3, med 1
qwen3-8B no-think	40	40	1.000	mean 297
qwen3-14B no-think	40	40	1.000	mean 115, med 45
qwen3-32B no-think	40	40	1.000	mean 102, med 55
qwen3-8B think 8K cap	8 answered	8	1.000	mean 531

own sums 26765 and 5782). One further local chain emitted an unevaluated expression at the derive step and was recorded as an honest abort, excluded from scoring.

Verifier	scale2	Ladder	Combined
Judge L (verdicts)	3/4	12/12	15/16
Judge M (verdicts)	4/4	12/12	16/16
Judge S (verdicts)	4/4	12/12	16/16
Cross-judge vote	1/4 (3 abstain)	12/12	13/16

Chain	Count	Sum	Derived	Outcome
frontier small	278	39081	39359	halted at count
frontier large	273	38808	39081	clean, accepted
frontier mid	273	38808	39081	clean, accepted
qwen3-14B (1)	453	9848	10301	halted at count
qwen3-14B (2)	221	6884	7105	halted at count
qwen3-32B (1)	254	8934	9188	halted at count
qwen3-32B (2)	265	26500	29150	halted at count
qwen3-8B (2)	119	5663	12862	halted at count